

IDS Fundamentals - TryHackMe

Overview

Completed the IDS Fundamentals room on TryHackMe to understand how Intrusion Detection Systems monitor network traffic , detect malicious activity, and generate security alert using Snort.

Learning Objectives

- Understand Intrusion Detection Systems (IDS)
- Difference between IDS and Firewall
- Anomaly-based Detection
- Network IDS (NIDS)
- Host IDS (HIDS)
- Introduction to Snort
- Analysing PCAP files using Snort

What is IDS ?

An Intrusion Detection System(IDS) monitors network or host activities and generates alerts when suspicious activity is detected. Unlike a firewall , an IDS detects and alerts but does not block traffic.

IDS VS Firewall

Firewall	IDS
Filters traffic	Monitors traffic
Allow/Block traffic	Generates alert
Prevention	Detective

Types of IDS

Network IDS (NIDS)

- Monitors networks traffic
- Detect attack across the network

Host IDS (HIDS)

- Installed on individual systems
- Monitors files, logs, and process

Detection Methods

Signature-based Detection

- ❖ Detects known attack using predefined rules.
- ❖ Fast and accurate for known threats.

Anomaly-based Detection

- ❖ Detects unusual behavior.
- ❖ Can identify previously unknown attacks.

Introduction to Snort

Snort is an open-source Network Intrusion Detection System (NIDS).

Functions:

- Packet Inspection
- Signature Matching
- Alert Generation
- PCAP Analysis

Snort Commands Used

Snort -V is used for check version of snort

Analyze PCAP (Packet Capture)

```
sudo snort -q -l /var/log/snort \
```

```
-r Intro_to_IDS.pcap \
```

```
-A alert_fast \
```

```
-c /etc/snort/snort.lua
```

Practical Exercise

Analysed the provided PCAP file using Snort.

Detected:

- SSH Connection Attempt
- Source IP Address
- Alert Message
- Rule SID

Tools Used

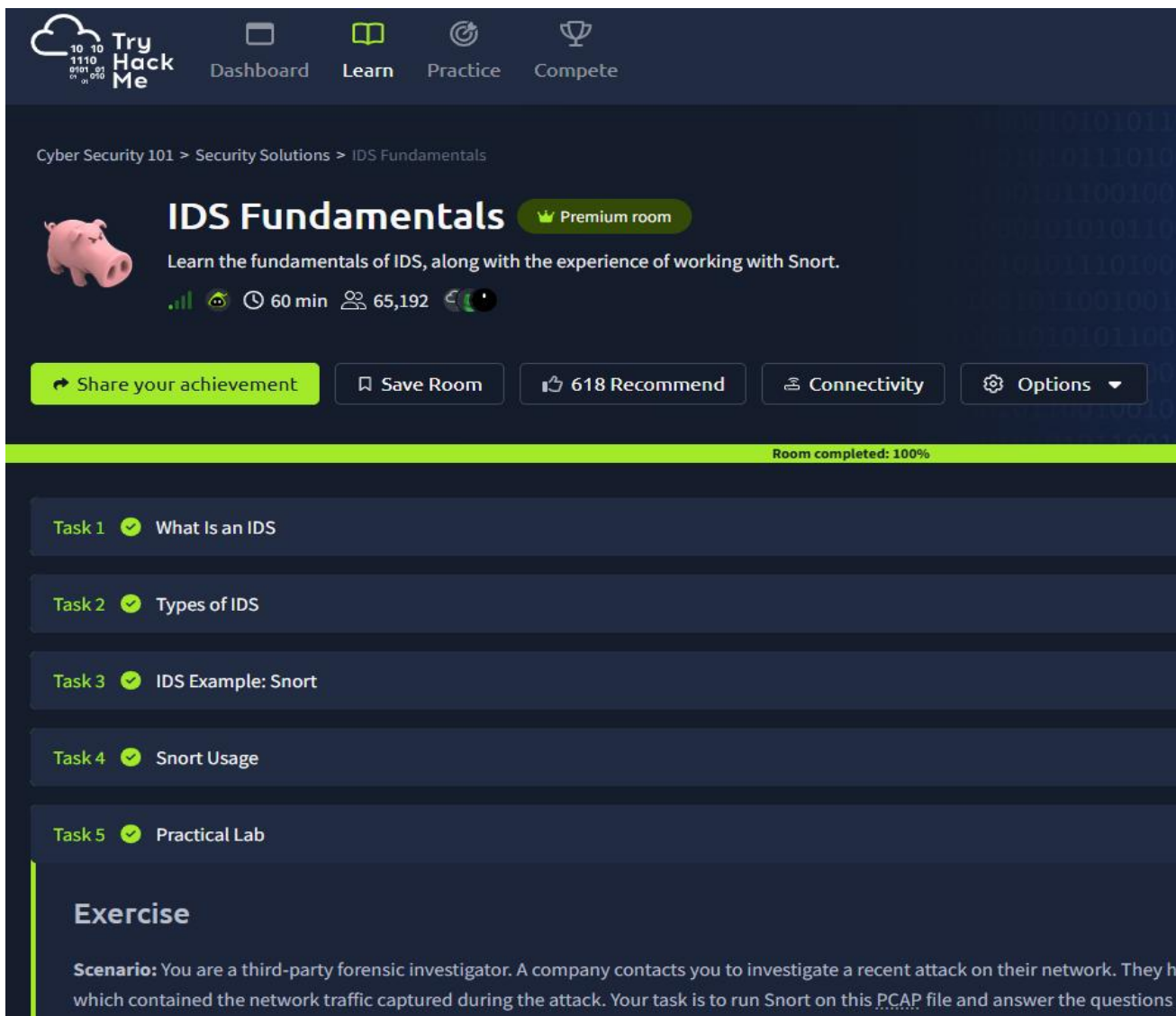
- TryHackMe
- Snort 3
- Linux Terminal
- PCAP Files

Skills Developed

- IDS Fundamentals
- Snort Basics
- PCAP Analysis
- Network Traffic Monitoring
- Signature-based Detection
- SOC Analytics Fundamentals

Link for Reference :

https://tryhackme.com/room/idsfundamentals?utm_campaign=social_share&utm_medium=social&utm_content=share-completed-room&utm_source=copy&shareId=69fb78461a6d745f64fa947f



The screenshot displays the TryHackMe interface for the 'IDS Fundamentals' room. The top navigation bar includes links to Dashboard, Learn, Practice, and Compete. The room title 'IDS Fundamentals' is prominently displayed with a 'Premium room' badge. A progress bar at the top indicates 'Room completed: 100%'. Below this, a list of five tasks is shown, each with a green checkmark indicating completion:

- Task 1: What Is an IDS
- Task 2: Types of IDS
- Task 3: IDS Example: Snort
- Task 4: Snort Usage
- Task 5: Practical Lab

The 'Exercise' section is visible at the bottom, starting with the following scenario:

Scenario: You are a third-party forensic investigator. A company contacts you to investigate a recent attack on their network. They have provided a PCAP file which contained the network traffic captured during the attack. Your task is to run Snort on this PCAP file and answer the questions...

